Brownian movement enables us,

I. to calculate Avogadro number.

II. to confirm kinetic theory which considers the ceaseless rapid movement of molecules that increases with increase in temperature.

III. to understand the stability of colloids: As the particles in continuous rapid movement they do not come close and hence not get condensed. That is Brownian movement does not allow the particles to be acted on by force of gravity.

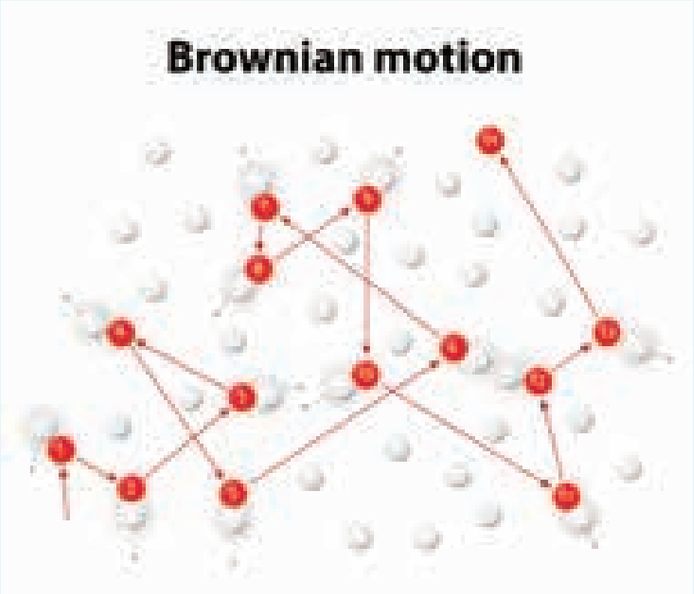


Figure 10. 13 Brownian movement

12) Electrical property

(1) Helmholtz double layer

The surface of colloidal particle adsorbs one type of ion due to preferential adsorption. This layer attracts the oppositely charged ions in the medium and hence at the boundary separating the two electrical double layers are setup. This is called as Helmholtz electrical double layer.

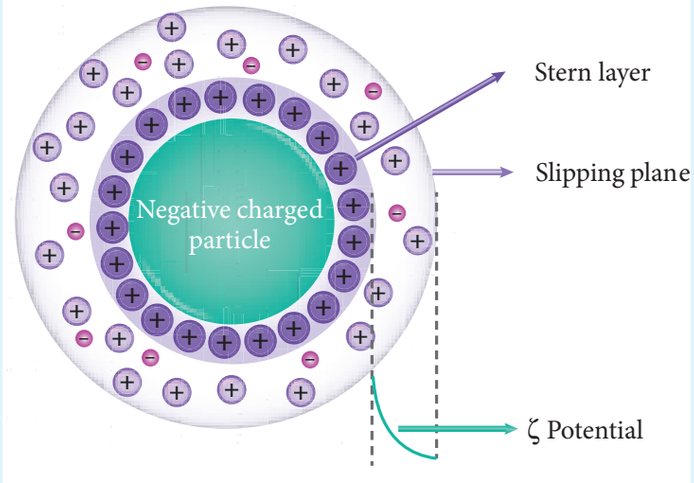


Figure 10.14 Electrical double layer

As the particles nearby are having similar charges, they cannot come close and condense. Hence this helps to explain the stability of a colloid.

(ii) Electrophoresis:

When electric potential is applied across two platinum electrodes dipped in a hydrophilic sol, the dispersed particles move toward one or other electrode.

This migration of sol particles under the influence of electric field is called electrophoresis or cataphoresis. If the sol particles migrate to the cathode, then they posses positive (+) charges,

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